

LASQ versus the Diameter Two Property

Verification of the fixed- ε obstruction, further obstructions, and quantitative partial progress

Research note prepared from the supplied source paper and partial proof

22 June 2026

Abstract

Rodríguez and Rueda Zoca asked whether a locally almost square (LASQ) Banach space can fail the diameter two property (D2P). The supplied partial proof concerns the fixed- ε renorming used in their construction of spaces with the slice-D2P but with arbitrarily small relatively weakly open subsets. That proof is correct. We sharpen it in three ways: a symmetric “leakage” inequality is obtained; every tree generator, not only the root, is shown to be quantitatively strongly extreme; and the actual norm on the whole space constructed in the proof of their Theorem 1.2 is shown not to be LASQ.

A literature search through 21 June 2026 did not locate a proof or counterexample for the general LASQ-versus-D2P question. We therefore examine two later natural candidates. The 2024 renorming of $L_\infty[0, 1]$ by Cobollo et al. and the 2026 dual LD2P-without-D2P example of Haller–Kaasik–Ostrak both contain explicit strongly extreme sphere points and hence are not LASQ. Finally, we prove a finite-rank estimate: if X is LASQ and a basic relatively weakly open subset of B_X is determined by an operator of rank r , then its diameter is at least $2/(r + 1)$. This does not settle the problem, but it isolates the remaining obstruction: LASQ supplies almost square directions at one point, while D2P requires such directions to be simultaneously small under finitely many functionals.

Contents

1	The question and the outcome of this investigation	2
2	The fixed-epsilon construction from the source paper	3
2.1	A symmetric leakage estimate	4
2.2	All tree generators are strongly extreme	4
3	The full norm used in the source theorem is not LASQ	5
4	Two later candidate constructions are also not LASQ	6
4.1	A general saturation obstruction in supremum norms	6
4.2	The 2024 L-infinity renorming	7
4.3	The 2026 dual LD2P-without-D2P example	7

5	The weakly-null-net route to D2P	8
6	A finite-rank lower bound for weakly open sets	9
7	Literature search and current status	10
8	Strategies tested and the remaining bottleneck	11
9	Evidence and packet history	11
10	Conclusion	12

1 The question and the outcome of this investigation

For a real Banach space X , write B_X and S_X for its closed unit ball and unit sphere. We use the following equivalent ε -form of local almost squareness:

$$X \text{ is LASQ} \iff \forall x \in S_X \forall \varepsilon > 0 \exists y \in S_X : \|x \pm y\| \leq 1 + \varepsilon.$$

The diameter two property (D2P) means that every non-empty relatively weakly open subset of B_X has diameter two. LASQ always implies the slice diameter two property (also called LD2P), whereas weak almost squareness implies D2P; see [1]. The question raised in [8] is:

Question 1.1 (Rodríguez-Rueda Zoca). *Does there exist a LASQ Banach space which fails the D2P?*

The source passage is recorded below from the arXiv PDF.

have been widely studied in the literature (see, e.g., [6, 4, 18]).

The techniques of Theorem 1.1 allow us to show the Lebesgue-Bochner space $L_1(\mu, Y)$ is WASQ for any Banach space Y whenever μ is a non-atomic finite measure (Corollary 2.5). This result should be compared with the above mentioned result of [27] that the property of being LASQ passes from a Banach space Y to the Köthe-Bochner space $E(Y)$, for any Banach function space E . We finish Section 2 with an example of a WASQ Banach space of the form $L_1(\nu)$ as in Theorem 1.1 which is not an $\mathcal{L}_1$ -space (Subsection 2.3).

In Section 3 we go a bit further in the analysis of the link between almost squareness and diameter two properties. One of the main questions raised in [2] is whether there exists a LASQ Banach space which is not WASQ. Very recently, Kaasik and Veeorg proved in [29, Section 2] that the answer is negative and that an example can be found in the class of Lipschitz-free spaces over complete metric spaces. For such spaces, the properties SD2P, D2P, slice-D2P and LASQ are equivalent (combine [9, Theorem 1.5] and [24, Theorem 3.1]), so the above mentioned example satisfies the SD2P. Since the slice-D2P and the D2P are different properties [10], it is a natural question whether there exists a LASQ Banach space which fails the D2P. Within the framework of Banach lattices, a stronger version of the LASQ property which implies the D2P has been considered in [16]. Even though we do not know the answer to the previous question, we make some progress in this direction. Our main result in Section 3 is the following:

The conclusions reached here are as follows.

1. The supplied fixed- ε obstruction is valid, and its quantitative estimate can be improved.
2. The entire norm constructed in the proof of [8, Theorem 1.2], not merely its c_0 core, is not LASQ.

3. No full resolution of the general question was found in the literature search described in section 7.
4. Two later constructions which at first sight are plausible sources of a counterexample are also not LASQ.
5. A general finite-rank lower bound, theorem 6.2, gives some positive information but falls short of diameter two.

A point $p \in S_X$ is *strongly extreme* if

$$\|p + y_n\| \longrightarrow 1, \quad \|p - y_n\| \longrightarrow 1 \implies \|y_n\| \longrightarrow 0.$$

Plainly, a LASQ space has no strongly extreme point in its unit sphere.

2 The fixed- ε construction from the source paper

The construction audited here is the fixed- ε renorming used in the proof of the source Theorem 1.2, shown in the following source crop.

4

J. RODRÍGUEZ AND A. RUEDA ZOCA

Theorem 1.2. *Let X be a Banach space containing a complemented isomorphic copy of c_0 . Then for any $0 < \varepsilon < 1$ there exists an equivalent norm $|\cdot|$ on X such that:*

- (i) $(X, |\cdot|)$ has the slice-D2P, that is, every slice of $B_{(X, |\cdot|)}$ has diameter 2.
- (ii) There are non-empty relatively weakly open subsets of $B_{(X, |\cdot|)}$ of arbitrarily small diameter.
- (iii) $(X, |\cdot|)$ is (r, s) -SQ for all $0 < r, s < \frac{1-\varepsilon}{1+\varepsilon}$ in the sense of [8, Section 6], that is, for every finite set $\{x_1, \dots, x_n\} \subseteq S_X$ there exists $y \in S_X$ satisfying

$$|rx_i \pm sy| \leq 1 \quad \text{for every } i \in \{1, \dots, n\}.$$

Condition (iii) measures somehow how far is the norm from being ASQ. Notice that a Banach space is ASQ if and only if it is (r, s) -SQ for all $0 < r, s < 1$.

Theorem 1.2 applies to any separable Banach space containing an isomorphic copy of c_0 , thanks to Sobczyk's theorem. The proof of Theorem 1.2 is inspired by the renorming technique developed by Becerra Guerrero, López-Perez and Rueda Zoca in [10, Theorem 2.4], which in turn uses ideas of the example of Argyros, Odell and Rosenthal [5] of a closed bounded convex subset of c_0 having the convex point

We retain the notation of [8, Section 3]. Let $\mathbb{N}^{<\omega}$ be the Baire tree and let

$$Z = c(\mathbb{N}^{<\omega}) \oplus_\infty c_0, \quad \|(x, y)\|_Z = \max\{\|x\|_\infty, \|y\|_\infty\}.$$

For $\alpha \in \mathbb{N}^{<\omega}$ put

$$x_\alpha(\beta) = \begin{cases} 1, & \beta \preceq \alpha, \\ -1, & \text{otherwise,} \end{cases} \quad A = \{x_\alpha : \alpha \in \mathbb{N}^{<\omega}\}.$$

Every x_α is $\{\pm 1\}$ -valued and $\lim x_\alpha = -1$. Fix $0 < \varepsilon < 1$ and set

$$B_\varepsilon = \overline{\text{co}}\left((A \times \{0\}) \cup (-A \times \{0\}) \cup ((1 - \varepsilon)B_Z + \varepsilon B_{c_0(\mathbb{N}^{<\omega})} \times \{0\})\right). \quad (1)$$

Let $|\cdot|_\varepsilon$ be the Minkowski functional of B_ε . The source proof gives

$$(1 - \varepsilon)B_Z \subset B_\varepsilon \subset B_Z, \quad \|z\|_Z \leq |z|_\varepsilon \leq \frac{1}{1 - \varepsilon} \|z\|_Z. \quad (2)$$

Let $P_2(x, y) = y$ and $L(x, y) = \lim x$.

2.1 A symmetric leakage estimate

The partial proof used the one-sided estimate

$$\|P_2 w\|_\infty \leq \frac{1-\varepsilon}{\varepsilon}(1+L(w)).$$

It is correct. In fact, the following symmetric inequality is available.

Lemma 2.1 (Leakage inequality). *For every $w \in B_\varepsilon$,*

$$\varepsilon \|P_2 w\|_\infty + (1-\varepsilon) |L(w)| \leq 1-\varepsilon. \quad (3)$$

Consequently,

$$\|P_2 w\|_\infty \leq \frac{1-\varepsilon}{\varepsilon}(1+L(w)) \quad \text{and} \quad \|P_2 w\|_\infty \leq \frac{1-\varepsilon}{\varepsilon}(1-L(w)). \quad (4)$$

Proof. The left side of (3) is a continuous convex function of w . It is therefore enough to check the three generating pieces in (1).

If $w = (\pm x_\alpha, 0)$, then $P_2 w = 0$ and $|L(w)| = 1$, so equality holds. If

$$w = (1-\varepsilon)u + \varepsilon(v, 0), \quad u \in B_Z, \quad v \in B_{c_0(\mathbb{N}^{<\omega})},$$

then $\|P_2 w\|_\infty \leq 1-\varepsilon$ and $|L(w)| \leq 1-\varepsilon$. Hence

$$\varepsilon \|P_2 w\|_\infty + (1-\varepsilon) |L(w)| \leq \varepsilon(1-\varepsilon) + (1-\varepsilon)^2 = 1-\varepsilon.$$

The closed convex hull is therefore contained in the sublevel set. The two estimates in (4) follow from $1-|t| \leq 1 \pm t$. $\square$

The geometric meaning is simple: the only generators which can carry a second c_0 coordinate have $|L| \leq 1-\varepsilon$. Thus approaching either face $L = \pm 1$ forces the second coordinate to vanish quantitatively.

2.2 All tree generators are strongly extreme

Theorem 2.2 (Quantitative strong extremity). *For every $\alpha \in \mathbb{N}^{<\omega}$, the points*

$$p_\alpha = (x_\alpha, 0) \quad \text{and} \quad -p_\alpha$$

are strongly extreme points of B_ε . More precisely, if $y = (u, v) \in Z$ and

$$|p_\alpha + y|_\varepsilon \leq 1 + \delta, \quad |p_\alpha - y|_\varepsilon \leq 1 + \delta \quad (5)$$

for some $\delta \geq 0$, then

$$|y|_\varepsilon \leq C_\varepsilon \delta, \quad C_\varepsilon = \max \left\{ \frac{1}{1-\varepsilon}, \frac{1}{\varepsilon} \right\}. \quad (6)$$

Proof. Because $B_\varepsilon \subset B_Z$, (5) implies

$$\|x_\alpha \pm u\|_\infty \leq 1 + \delta.$$

Every coordinate of x_α has modulus one. For a scalar $a \in \{\pm 1\}$,

$$\max\{|a+t|, |a-t|\} = 1 + |t|.$$

It follows coordinatewise that

$$\|u\|_\infty \leq \delta, \quad |\lim u| \leq \delta. \quad (7)$$

Put $\ell = \lim u$ and

$$q_+ = \frac{p_\alpha + y}{1 + \delta}, \quad q_- = \frac{p_\alpha - y}{1 + \delta}.$$

Both points lie in B_ε . Applying the first inequality in (4) to q_+ and q_- gives, respectively,

$$\|v\|_\infty \leq \frac{1-\varepsilon}{\varepsilon}(\delta + \ell), \quad \|v\|_\infty \leq \frac{1-\varepsilon}{\varepsilon}(\delta - \ell).$$

Therefore

$$\|v\|_\infty \leq \frac{1-\varepsilon}{\varepsilon}(\delta - |\ell|) \leq \frac{1-\varepsilon}{\varepsilon}\delta. \quad (8)$$

Combining (7), (8), and (2),

$$|y|_\varepsilon \leq \frac{1}{1-\varepsilon} \max\left\{\delta, \frac{1-\varepsilon}{\varepsilon}\delta\right\} = C_\varepsilon \delta.$$

This proves the quantitative estimate and hence strong extremity. The assertion for $-p_\alpha$ follows by symmetry. $\square$

Corollary 2.3. *The partial result supplied with the problem is correct. Its factor $2/\varepsilon$ can be replaced by $1/\varepsilon$, and its conclusion holds at every tree point $\pm(x_\alpha, 0)$.*

Proof. If $|p_\alpha \pm y_n|_\varepsilon \rightarrow 1$, then

$$\delta_n = \max\{|p_\alpha + y_n|_\varepsilon, |p_\alpha - y_n|_\varepsilon\} - 1$$

is non-negative by the triangle inequality and tends to zero. Apply (6). Since LASQ would require vectors of norm tending to one at every sphere point, the space is not LASQ. $\square$

3 The full norm used in the source theorem is not LASQ

The source proof first constructs the preceding norm on a space isomorphic to c_0 . For a general space containing a complemented copy of c_0 , it then writes $X = E \oplus W$ and equips X with an ℓ_∞ -sum norm. The following elementary fact closes the gap between the core obstruction and the full construction.

Lemma 3.1 (ℓ_∞ sums). *For non-zero Banach spaces X and Y ,*

$$X \oplus_\infty Y \text{ is LASQ} \iff X \text{ is LASQ or } Y \text{ is LASQ}.$$

Proof. Suppose first that X is LASQ. Given $(x, y) \in S_{X \oplus_\infty Y}$, choose $u_n \in S_X$ with

$$\|x/\|x\| \pm u_n\| \rightarrow 1$$

when $x \neq 0$; if $x = 0$, take any $u_n \in S_X$. Convexity gives

$$\limsup_n \|x \pm u_n\| \leq 1.$$

Consequently,

$$\|(x, y) \pm (u_n, 0)\| = \max\{\|x \pm u_n\|, \|y\|\} \longrightarrow 1.$$

The same argument applies if Y is LASQ.

Conversely, suppose $X \oplus_\infty Y$ is LASQ and take $x \in S_X, y \in S_Y$. There are $(u_n, v_n) \in B_{X \oplus_\infty Y}$ such that

$$\max\{\|u_n\|, \|v_n\|\} \rightarrow 1$$

and

$$\max\{\|x \pm u_n\|, \|y \pm v_n\|\} \rightarrow 1.$$

After passing to a subsequence, either $\|u_n\| \rightarrow 1$ or $\|v_n\| \rightarrow 1$. In the first case x has an LASQ sequence in X ; in the second case y has one in Y . If neither summand were LASQ, choosing x and y to be bad sphere points would give a contradiction. $\square$

Theorem 3.2. *The particular equivalent norm on X constructed in the proof of [8, Theorem 1.2] is not LASQ for any fixed $0 < \varepsilon < 1$.*

Proof. The c_0 summand E is equipped with the fixed- ε norm of section 2, so it is not LASQ by theorem 2.2. If the complementary summand W is non-zero, the source proof chooses on W an equivalent norm whose unit ball has slices of arbitrarily small diameter. Such a space cannot be LASQ, because LASQ implies the slice-D2P. The final norm is the ℓ_∞ -sum of these two norms, so theorem 3.1 applies. If $W = \{0\}$, the conclusion is already theorem 2.2. $\square$

Remark 3.3. *The conclusion concerns the explicit norm built in the proof of the theorem. It does not say that every norm satisfying properties (i)–(iii) of [8, Theorem 1.2] must fail LASQ.*

4 Two later candidate constructions are also not LASQ

4.1 A general saturation obstruction in supremum norms

Lemma 4.1 (Supremum-norm saturation). *Let E be a subspace of $\ell_\infty(\Gamma)$ or $L_\infty(\mu)$, and let $\|\cdot\|$ be an equivalent norm satisfying*

$$\|z\|_\infty \leq \|z\| \leq C \|z\|_\infty \quad (z \in E). \quad (9)$$

If $u \in S_{(E, \|\cdot\|)}$ is $\{\pm 1\}$ -valued pointwise (or almost everywhere), then u is strongly extreme. Quantitatively,

$$\|u \pm z\| \leq 1 + \delta \implies \|z\| \leq C\delta.$$

Proof. The left inequality in (9) yields $\|u \pm z\|_\infty \leq 1 + \delta$. Pointwise,

$$\max\{|u(t) + z(t)|, |u(t) - z(t)|\} = 1 + |z(t)|,$$

so $\|z\|_\infty \leq \delta$. The right inequality in (9) finishes the proof. $\square$

4.2 The 2024 L_∞ renorming

Cobollo, Isert, López-Pérez, Martín, Perreau, Quero, Quilis, Rodríguez-Vidanes and Rueda Zoca constructed in [3] an equivalent norm on $L_\infty[0, 1]$ whose unit ball has relatively weakly open subsets of arbitrarily small diameter and a weakly dense set of Daugavet points. Their unit ball has the form

$$\mathcal{B}_\varepsilon = \overline{\text{co}}\left((2K_0 - \mathbf{1}) \cup (-2K_0 + \mathbf{1}) \cup ((1 - \varepsilon)B_{L_\infty} + \varepsilon B_{\ker E_0})\right),$$

with

$$(1 - \varepsilon)B_{L_\infty} \subset \mathcal{B}_\varepsilon \subset B_{L_\infty}.$$

Their tree functions $u_\alpha = 2h_\alpha - \mathbf{1}$ belong to the new unit sphere and are $\{\pm 1\}$ -valued almost everywhere.

Corollary 4.2. *For every fixed $0 < \varepsilon < 1$, the norm of [3, Theorem 1.3] is not LASQ. Every tree function u_α is strongly extreme, with*

$$\|g\| \leq \frac{\delta}{1 - \varepsilon} \quad \text{whenever} \quad \|u_\alpha \pm g\| \leq 1 + \delta.$$

Proof. The norm comparison is exactly (9) with $C = (1 - \varepsilon)^{-1}$. Apply theorem 4.1. $\square$

The cited paper proves that the same u_α are preserved extreme Daugavet points. Strong extremity is a different and stronger local obstruction for the present purpose; it appears to follow immediately from the ambient L_∞ saturation but was not used there.

4.3 The 2026 dual LD2P-without-D2P example

Haller, Kaasik and Ostrak constructed a metric space M for which $\text{Lip}_0(M)$ has the LD2P but fails even the weak-star D2P [6, Example 5.2]. Its points are

$$M = \{x_i, y_i, u_j^i, v_j^i : i \in \{1, 2, 3\}, j \in \mathbb{N}\}.$$

The pairs at distance one are

$$y_1x_2, \quad y_2x_3, \quad y_3x_1,$$

and, for each i, j , the three edges in the path

$$x_i - u_j^i - v_j^i - y_i.$$

Every other pair of distinct points has distance two. Take x_1 as base point.

The distance-one graph is bipartite, with colour classes

$$\mathcal{C}_0 = \{x_i, v_j^i\}, \quad \mathcal{C}_1 = \{y_i, u_j^i\}.$$

Define $f \in \text{Lip}_0(M)$ by $f = 0$ on $\mathcal{C}_0$ and $f = 1$ on $\mathcal{C}_1$.

Proposition 4.3. *The function f is a strongly extreme point of $B_{\text{Lip}_0(M)}$. More precisely,*

$$\|f \pm g\|_{\text{Lip}} \leq 1 + \delta \implies \|g\|_{\text{Lip}} \leq 3\delta.$$

Consequently the space in [6, Example 5.2] is not LASQ.

Proof. The function f changes by exactly one across every distance-one edge, hence $\|f\|_{\text{Lip}} = 1$. If p, q form such an edge and $a = g(p) - g(q)$, the two assumptions give

$$|1 + a| \leq 1 + \delta, \quad |1 - a| \leq 1 + \delta,$$

after possibly reversing the orientation. Thus $|a| \leq \delta$.

The distance-one graph has graph diameter six. Hence any two vertices can be joined by at most six unit edges, and therefore

$$|g(p) - g(q)| \leq 6\delta \quad (p, q \in M).$$

On a distance-one pair the Lipschitz quotient is at most δ ; on every other pair the metric distance is two, so the quotient is at most 3δ . This proves the estimate. $\square$

Thus the most recent explicit dual example separating LD2P from D2P does not answer the LASQ question either.

5 The weakly-null-net route to D2P

The central missing ingredient can be formulated cleanly. Say that X has the *net-LASQ property* if for every $x \in S_X$ there is a weakly null net (y_γ) in X such that

$$\|y_\gamma\| \rightarrow 1, \quad \|x \pm y_\gamma\| \rightarrow 1.$$

Proposition 5.1. *Every Banach space with the net-LASQ property has the D2P.*

Proof. Let $U \subset B_X$ be non-empty and relatively weakly open. Since a net-LASQ space is infinite-dimensional, S_X is weakly dense in B_X ; choose $x \in U \cap S_X$. There are $f_1, \dots, f_n \in X^*$ and $\rho > 0$ such that

$$W = \{z \in B_X : |f_i(z - x)| < \rho \ (i = 1, \dots, n)\} \subset U.$$

Let (y_γ) be a net as above and set

$$\eta_\gamma = \max\{\|x + y_\gamma\|, \|x - y_\gamma\|, 1\} - 1.$$

Then $\eta_\gamma \rightarrow 0$ and

$$z_\gamma^\pm = \frac{x \pm y_\gamma}{1 + \eta_\gamma} \in B_X.$$

Since $y_\gamma \rightarrow 0$ weakly,

$$f_i(z_\gamma^\pm - x) = \frac{\pm f_i(y_\gamma) - \eta_\gamma f_i(x)}{1 + \eta_\gamma} \rightarrow 0.$$

Thus $z_\gamma^\pm \in W$ eventually, while

$$\|z_\gamma^+ - z_\gamma^-\| = \frac{2\|y_\gamma\|}{1 + \eta_\gamma} \rightarrow 2.$$

Hence $\text{diam } U = 2$. $\square$

Ciacci proved that positive LASQ Banach lattices have the net-LASQ property, and consequently the D2P [2]. The same paper explicitly asks whether ordinary LASQ implies this net property. A positive solution of that stronger question would settle the original LASQ-versus-D2P problem positively.

6 A finite-rank lower bound for weakly open sets

The next result is independent of the special constructions above. It quantifies what LASQ alone forces inside a weak neighborhood described by finitely many functionals.

Lemma 6.1 (Almost diametral pairs over extreme quotient points). *Let X be LASQ, let $T : X \rightarrow F$ be a finite-rank operator, and put*

$$K = \overline{T(B_X)} \subset F.$$

If $v \in \text{ext } K$, then for every $\eta > 0$ there are $a, b \in B_X$ such that

$$\|Ta - v\| < \eta, \quad \|Tb - v\| < \eta, \quad \|a - b\| > 2 - \eta.$$

Proof. If $T = 0$ there is nothing to prove, so assume K is non-trivial. Since K is symmetric, 0 is not an extreme point. Choose $z_k \in B_X$ with $Tz_k \rightarrow v$. We claim that $\|z_k\| \rightarrow 1$. Otherwise, after passing to a subsequence, $\|z_k\| \leq c < 1$. Then $v \in cK$, so $v = cw + (1 - c)0$ for some $w \in K$, contradicting extremality.

Put $x_k = z_k / \|z_k\| \in S_X$, so $Tx_k \rightarrow v$. By LASQ choose $y_k \in S_X$ and numbers $\varepsilon_k \downarrow 0$ such that

$$\|x_k \pm y_k\| \leq 1 + \varepsilon_k.$$

After passing to a subsequence, $Ty_k \rightarrow h \in F$. Set

$$a_k = \frac{x_k + y_k}{1 + \varepsilon_k}, \quad b_k = \frac{x_k - y_k}{1 + \varepsilon_k}.$$

Then $a_k, b_k \in B_X$, $\|a_k - b_k\| = 2/(1 + \varepsilon_k) \rightarrow 2$, and

$$Ta_k \rightarrow v + h, \quad Tb_k \rightarrow v - h.$$

Both limits belong to K . Since v is their midpoint and is extreme, $h = 0$. The assertion follows for large k . $\square$

Theorem 6.2 (Finite-rank diameter estimate). *Let X be LASQ, let $T : X \rightarrow F$ have rank $r < \infty$, and let $V \subset T(X)$ be open. If*

$$U = B_X \cap T^{-1}(V) \neq \emptyset,$$

then

$$\text{diam } U \geq \frac{2}{r + 1}. \quad (10)$$

For $r = 0$, the right conclusion is understood as $\text{diam } U = 2$.

Proof. Assume $r \geq 1$ and put $K = \overline{T(B_X)}$. Choose $x_0 \in U$ and set $c = Tx_0 \in V \cap K$. By the finite-dimensional Minkowski–Carathéodory theorem, there are extreme points $v_1, \dots, v_m$ of K and coefficients $\lambda_i \geq 0$ such that

$$c = \sum_{i=1}^m \lambda_i v_i, \quad \sum_{i=1}^m \lambda_i = 1, \quad m \leq r + 1.$$

Choose j with $\lambda_j = \max_i \lambda_i$, so $\lambda_j \geq 1/(r + 1)$.

Fix $\eta > 0$. By theorem 6.1, choose $a_j, b_j \in B_X$ with Ta_j, Tb_j as close to v_j as desired and $\|a_j - b_j\| > 2 - \eta$. For $i \neq j$, choose $c_i \in B_X$ with Tc_i as close to v_i as desired. Define

$$a = \lambda_j a_j + \sum_{i \neq j} \lambda_i c_i, \quad b = \lambda_j b_j + \sum_{i \neq j} \lambda_i c_i.$$

Both points belong to B_X . Because V is open and Ta, Tb can be made arbitrarily close to c , the choices can be made so that $a, b \in U$. Finally,

$$\|a - b\| = \lambda_j \|a_j - b_j\| > \lambda_j(2 - \eta).$$

Letting $\eta \downarrow 0$ gives (10). □

Corollary 6.3. *If a basic relatively weakly open subset of B_X is described using r linearly independent functionals, then an LASQ space forces its diameter to be at least $2/(r + 1)$.*

Remark 6.4. *This estimate is compatible with the source construction. Its weakly open sets of small diameter use an increasing number of tree-coordinate constraints. The bound tends to zero as the number of independent functionals grows, so it cannot by itself prove D2P.*

7 Literature search and current status

The literature search covered the exact problem statement, the terms LASQ, WASQ, D2P, slice-D2P/LD2P, recent work on ultrapowers, Banach lattices, Lipschitz-free spaces, spaces of Lipschitz functions, and the renormings most closely related to the source construction. The following are the most relevant checkpoints.

Source	Result	Relevance here
Rodríguez–Rueda Zoca (2023) [8]	Poses the LASQ-without-D2P question and constructs the fixed- ε near-square renorming.	The explicit norm is ruled out in sections 2 and 3.
Ciaci (2023) [2]	Positive LASQ Banach lattices have a weakly null LASQ net and hence D2P.	Gives the strongest general positive mechanism currently found; asks whether the net conclusion holds for every LASQ space.
Rueda Zoca (2025) [9]	LASQ yields strong uniform slice-D2P consequences in ultrapowers.	Strengthens the slice conclusion, but does not establish D2P.
Haller–Kaasik–Ostrak (2023) and Kaasik (2026) [5, 7]	For Lipschitz-free spaces, LASQ is equivalent to the underlying complete metric space being a length space, and hence to Daugavet/D2P behaviour.	The problem has a positive answer in this special class.
Haller–Kaasik–Ostrak (2026) [6]	Constructs a dual space $\text{Lip}_0(M)$ with LD2P but not D2P.	The explicit checkerboard point in theorem 4.3 shows this example is not LASQ.
Cobollo et al. (2024) [3]	Renorms L_∞ with arbitrarily small weakly open subsets and many Daugavet points.	Its $\{\pm 1\}$ tree points are strongly extreme; see theorem 4.2.

No article, preprint, thesis, or survey located in this search states a general proof that LASQ implies D2P or gives a LASQ counterexample to D2P. The most recent sources surveyed continue to record only special classes and stronger auxiliary hypotheses. This is necessarily a literature-search conclusion rather than a proof of non-existence of an unindexed result.

8 Strategies tested and the remaining bottleneck

Several natural ways of turning the source’s “almost LASQ” norms into an exact LASQ norm do not produce a counterexample.

Letting $\varepsilon \downarrow 0$ on the same space. If one takes the closed convex hull of the union of source balls B_{ε_n} with $\varepsilon_n \downarrow 0$, then the inclusions $(1 - \varepsilon_n)B_Z \subset B_{\varepsilon_n} \subset B_Z$ force the resulting ball to be exactly B_Z . The limiting space $c(\mathbb{N}^{<\omega}) \oplus_{\infty} c_0$ is ASQ because of its c_0 summand, and therefore has the D2P.

Ultraproducts of the fixed- ε_n norms. The source spaces are (r, s) -SQ for all $r, s < (1 - \varepsilon_n)/(1 + \varepsilon_n)$. When $\varepsilon_n \rightarrow 0$, these constants tend to one, and the standard ultraproduct argument upgrades the limit to ASQ. Again the D2P is recovered rather than destroyed.

Direct sums. A c_0 -sum creates tail directions and is ASQ. An ℓ_{∞} -sum is LASQ only when at least one factor is LASQ, by theorem 3.1; if that factor has D2P, the usual product weak neighborhoods retain diameter two. Other standard absolute sums do not bypass the same compatibility constraints; see [4].

Finite-dimensional images. The proof of theorem 6.2 shows that over an extreme point of the finite-dimensional image $\overline{T(B_X)}$, LASQ directions must become small under T . At an interior image point, however, a non-zero symmetric segment can remain inside the image. Carathéodory decomposition then loses a factor of at most $r + 1$. Eliminating that loss would amount to finding one almost-square direction which is small under all the defining functionals, precisely the net-LASQ issue of section 5.

Thus the fixed- ε leakage mechanism is now well understood, but no construction found here simultaneously achieves the following two incompatible-looking requirements:

- at every sphere point, a norm-one almost-square direction; and
- in one finite-codimensional weak neighborhood, a uniform obstruction to all almost diametral pairs.

9 Evidence and packet history

This packet supersedes the June 15, 2026 root-point obstruction packet. That earlier packet proved that the root generator in the source fixed- ε renorming is strongly extreme. The active packet keeps the same source target but incorporates the supplied June 22 research note, which strengthens the obstruction to all tree generators, the full norm used in the source proof, two later candidate constructions, and a finite-rank lower bound for weakly open subsets of LASQ unit balls. The general LASQ-versus-D2P question remains open.

10 Conclusion

The supplied partial proof is mathematically sound. The fixed- ε source ball has a sharper symmetric leakage inequality, and all of its tree generators are quantitatively strongly extreme. This also rules out the complete norm actually used in the source proof of Theorem 1.2. Two prominent later LD2P-without-D2P constructions are ruled out by similarly explicit strong-extremity arguments.

The general LASQ-versus-D2P question was not resolved by the literature search or by the additional arguments developed here. The finite-rank bound $2/(r+1)$ is a rigorous partial result, while the weakly-null-net formulation identifies a clean sufficient condition whose validity for arbitrary LASQ spaces remains the central bottleneck.

References

- [1] T. A. Abrahamsen, J. Langemets and V. Lima, *Almost square Banach spaces*, J. Math. Anal. Appl. **434** (2016), 1549–1565. doi:10.1016/j.jmaa.2015.10.013.
- [2] S. Ciaci, *Locally almost square Banach lattices*, Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat. RACSAM **117** (2023), article 97. doi:10.1007/s13398-023-01434-6.
- [3] C. Cobollo, D. Isert, G. López-Pérez, M. Martín, Y. Perreau, A. Quero, A. Quilis, D. L. Rodríguez-Vidanes and A. Rueda Zoca, *Banach spaces with small weakly open subsets of the unit ball and massive sets of Daugavet and Δ -points*, Rev. R. Acad. Cienc. Exactas Fís. Nat. Ser. A Mat. RACSAM **118** (2024), article 96. doi:10.1007/s13398-024-01596-x.
- [4] J.-D. Hardtke, *Summands in locally almost square and locally octahedral spaces*, Acta Comment. Univ. Tartu. Math. **22** (2018), 149–162. doi:10.12697/ACUTM.2018.22.13.
- [5] R. Haller, J. K. Kaasik and A. Ostrak, *The Lipschitz-free space over a length space is locally almost square but never almost square*, Mediterr. J. Math. **20** (2023), article 39. doi:10.1007/s00009-022-02228-9.
- [6] R. Haller, J. K. Kaasik and A. Ostrak, *The local diameter two property and the diameter two property in spaces of Lipschitz functions*, J. Math. Anal. Appl. **556** (2026), article 130178. doi:10.1016/j.jmaa.2025.130178.
- [7] J. K. Kaasik, *Diameter two properties and almost square properties in Lipschitz-free spaces and spaces of Lipschitz functions*, Dissertationes Mathematicae Universitatis Tartuensis 146, University of Tartu Press, 2026.
- [8] J. Rodríguez and A. Rueda Zoca, *On weakly almost square Banach spaces*, Proc. Edinb. Math. Soc. **66** (2023), 979–997. doi:10.1017/S0013091523000536.
- [9] A. Rueda Zoca, *Slice diameter two property in ultrapowers*, Bull. Malays. Math. Sci. Soc. **48** (2025), article 62. doi:10.1007/s40840-025-01847-z.